

Yash Tandon

858-518-9839 | ytandon@ucsd.edu | [linkedin.com/in/yashtandon05/](https://www.linkedin.com/in/yashtandon05/) | github.com/yashtandon05

EDUCATION

University of California, San Diego

B.S. Data Science

B.S. Cognitive Science, Specialization in Machine Learning & Neural Computation

La Jolla, CA

Expected June 2027

Cumulative GPA: 4.0/4.0

PUBLICATIONS

IEEE Intelligent Transportation Systems Conference (ITSC)

- **Tandon, Y.**, Lopez, J.G.T., Blennemann, M., et al. "When Stopping Fails: Rethinking Minimal Risk Conditions through Human-Interactive Autonomous Driving for Safe Transportation Systems." *IEEE ITSC 2026*.
- Martinez-Sanchez, A., Ng, K., Maia, W., **Tandon, Y.**, et al. "Vision-Language Work Zone Intelligence for Safety-Critical Speed Regulation of Mixed-Autonomy Vehicles in Dynamic Environments." *IEEE ITSC 2026*.

EXPERIENCE

Laboratory for Intelligent & Safe Automobiles (LISA)

La Jolla, CA

Undergraduate Researcher | Collaborating with Mi3 Lab, UC Merced

July 2025 – Present

- Built real-time work-zone perception pipeline (YOLO + CLIP + EMA hysteresis state machine) achieving 96.5% event-level recall and 84.6% false positive reduction over baseline, deployed on NVIDIA Jetson Nano at 15 FPS.
- Developed a long-horizon instructional autonomous driving dataset by annotating nuPlan driving trajectories and leveraging vision-language models (LLaVA) to ground natural-language instructions into structured scene context.

Astera Holdings

New York, NY (Remote)

Quantitative Research Intern

June 2025 – September 2025

- Trained and calibrated supervised ML models (Logistic Regression, XGBoost, Random Forest) on 15,000+ football matches using 500+ engineered features, achieving 71% test accuracy in multi-class match outcome prediction.
- Deployed on AWS (S3, EC2) an options trading system for prediction markets (Kalshi) with WebSocket API integration; implemented backtesting engine for statistical arbitrage strategies (mean reversion via Z-score, RSI).

Data Science Alliance

San Diego, CA

Data Scientist Intern

June 2025 – August 2025

- Architected discrete-event simulation of homelessness systems modeling real-world inflows/outflows dynamics and resource constraints; derived min-cost funding strategy via TSPE optimization to achieve functional-zero in 5 years.
- Designed YOLO-based perception pipeline with temporal smoothing, reducing duplicate vehicle detections by 95%.
- Cleaned 200K+ row SQL data tables, eliminating missingness and enforcing schema constraints on confidential data.

Machine Learning Research Assistant

July 2024 – May 2025

- Conducted stakeholder interviews to identify real-world distribution factors; modeled optimal food bank locations via P-Median algorithm, cutting runtime by 50%, boosting accuracy by 15%, and increasing data granularity by 3x.

PROJECTS & LEADERSHIP

AI Team Lead for RoboCup SSL | PyTorch, Python, C++, OpenCV, UDP

October 2024 – Present

- Led team to first-ever World Championship qualification in the team's 5-year history, competing internationally.
- Trained deep reinforcement learning policies (JAL+HER) in PyTorch for automated, smart multi-agent coordination
- Modernized robot control via trajectory planning algorithms to boost Ball Placement success rate from 40% to 90%.

Sim Civics: Multiplayer Governance Simulator | Next.js, FastAPI, PostgreSQL, OpenAI API

May 2026

- Engineered LLM compiler pipeline with prompt chaining, abuse-detection guardrail, and structured JSON extraction
- Ran fairness evaluation across LLM input phrasings; reduced simulation output divergence from 21.84% to 5.57%.

Occluded Pedestrian Detection | Ultralytics, PyTorch, Python

January 2026 – March 2026

- Fine-tuned YOLOv8s with occlusion-aware data augmentation using structured black-mask & noise-based augmentations to simulate realistic obstruction; improved detection by 5% mAP@0.5 and 8% recall over baseline.

TECHNICAL SKILLS & AWARDS

Languages: Python, C/C++, Java, SQL, JavaScript, HTML/CSS

Tools & Libraries: PyTorch, Scikit-Learn, OpenCV, YOLO, Pandas, NumPy, AWS, Docker, Git, NVIDIA Alpamayo

Accolades: IMC Prosperity 3 (Top 6%) · Silver, IYMC 2021 (Top 10%) · Gold + Merit Award, UKMT Senior Maths Challenge 2021 (Top 2.5%) · CEMC Cayley 2021 & Pascal 2020 (Tied 4th Nationally) · Finalist, HiMCM 2020 (Top 7%)